

Program List for Practice

A. Preliminary

1. Write a program to print “**Hello World !**”.
2. Write a program to take input from the keyboard.
3. The length and height of a rectangle and radius of a circle are input through the keyboard. Write a program to find the area & perimeter of the rectangle, and the area & circumference of the circle.
4. Rahim’s basic salary is input through the keyboard. His house rent allowance is 30% of basic salary and medical allowance is 5% of basic salary. He gets an extra 1000 tk as technical allowance. Write a program to calculate his gross salary and print the result.
5. The distance between AUST main campus and Rajshahi campus (in km) is input through the keyboard. Write a program to convert and print this distance in meters, feet, inches, and centimeters.
6. The temperature of a city in Fahrenheit degrees is input through the keyboard. Write a program to convert this temperature into centigrade degrees.
7. Two numbers are input through the keyboard into two variables A and B. Write a program to interchange the contents of A and B.
8. If marks obtained by a student in 5 different subjects are input from the keyboard, find out the aggregate marks and percentage marks obtained by the student.
9. If a 5 digit number is input through the keyboard, write a program to calculate and print the sum of its digits. [Hint: Use the modulus operator ‘%’]
10. If a 5 digit number is input through the keyboard, write a program to reverse the number.
11. If a 4 digit number is input through the keyboard, write a program to obtain the sum of the first and last digit of this number.

B. Conditional

1. Two numbers are input through the keyboard. Write a program to find out the **maximum** and **minimum** of these 2 numbers.
2. Take a year as input and determine whether it is a **leap year**. [Hint: Check the divisibility by 4, 100, and 400]
3. If the cost price and selling price of an item is input through the keyboard, write a program to determine whether the seller has made profit or incurred loss. Also, determine how much profit he made or loss he incurred.
4. Any integer is input through the keyboard. Write a program to find out whether it is an **odd number** or **even number**.
5. According to the Gregorian calendar, it was Monday on the date 01/01/1900. If any year is input through the keyboard, write a program to find out what is the day on 1st January of that year.
6. A five digit number is entered through the keyboard. Write a program to obtain the reverse number and to determine whether the original and reverse numbers are equal or not.
7. AUST grading policy is:
 1. 80 % marks or above is A+
 2. 75% to 79% marks is A
 3. 70% to 74% marks is A-
 4. 65% to 69% marks is B+
 5. 60% to 64% marks is B
 6. 55% to 59% marks is B-
 7. 50% to 54% marks is C+
 8. 45% to 49% marks is C
 9. 40% to 44% marks is D
 10. Below 40% is F

Write a program which will take an input from the user and calculate the grade of a student according to AUST grading policy based on that input.

8. A certain grade of steel is determined according to the following conditions:
 1. Hardness must be greater than 60
 2. Carbon content must be less than 0.7

3. Tensile strength must be greater than 5000

The grades are as follows:

1. Grade is 10 if all three conditions are met
2. Grade is 9 if condition (i) and (ii) are met
3. Grade is 8 if condition (ii) and (iii) are met
4. Grade is 7 if condition (i) and (iii) are met
5. Grade is 6 if only one condition is met
6. Grade is 5 if none of the conditions are met

Write a program which will require the user to give values of hardness, carbon content, and tensile strength of the steel, and output the grade of the steel.

C. Lopping

1. x and n are taken as input from the user. Write a program to compute x^n and $n!$
2. Write a program to determine the **GCD** (greatest common divisor) and **LCM** (least common multiple) of 3 numbers.
3. Find out the sum of each of the following series. n is the input from the user.
 1. $3 + 11 + 19 + \dots + 1691$.
 2. $7 + 20 + 33 + \dots$ (up to 100 th term)
 3. $5 - 11 + 17 - \dots$ (up to 75 th term)
 4. $1 + (1 + 2) + (1 + 2 + 3) + \dots + (1 + 2 + 3 + \dots + n)$
 5. $2 * 7 * 12 * \dots * 37$
5. Write a program to determine all **prime numbers** within the range [a-b], where a & b are input through the keyboard.
6. Construct the following table. Here n is input from the user.

1	2	3		n
2	4	6		2n
3	6	9		3n
.....				
.....				
.....				
n	2n	3n	...	nn
7. Write a program to find out the first n **perfect number**, where n is the input from the user.
8. Write a program to find the first n **Fibonacci number**, where n is the input from the user.
9. Write a program to show the following triangle/rectangle of '*'s or numbers. Take n as input from the user to determine the number of rows of the structure. (e.g. n = 5)

```

*
* * *
* * * * *
* * * * *
* * * * *

```

```

*
* *
* *
* *
* * * * *

```

```

1
1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
1 2 3 4 5 4 3 2 1

```

```

* * * * *
* * * * *

```

* * * * *
* * * * *
* * * * *

* * * * *
* *
* *
* *
* * * * *

1 2 3 2 1
1 1
1 1
1 1
1 2 3 2 1

*
* * *
* * * * *
* * * * * * *
* * * * * * * * *
* * * * * * *
* * * * *
* * *
*

1
1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
1 2 3 4 5 4 3 2 1
1 2 3 4 3 2 1
1 2 3 2 1
1 2 1
1

* * * * * * * * *
* * * * * * *
* * * * *
* * *
*
* * *
* * * * *
* * * * * * *
* * * * * * * * *

1 2 3 4 5 4 3 2 1
1 2 3 4 3 2 1
1 2 3 2 1
1 2 1
1
1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
1 2 3 4 5 4 3 2 1

10. Write a program to print out all **Armstrong numbers** between 1 and 10000. If the sum of cubes of each digit of the number is equal to the number itself, then the number is called an Armstrong number. For example, $153 = (1*1*1) + (5*5*5) + (3*3*3)$.

D. Function

1. Write a function to calculate the **factorial** value of any integer entered through the keyboard.
2. Write a function **power(a,b)** to calculate the value of a raised to b.
3. Write a function to calculate **LCM** of two numbers.
4. Write a function to calculate **GCD** of two numbers.
5. Any year is entered through the keyboard. Write a function to determine whether the year is a leap year or not.
6. An integer is entered through the keyboard. Write a function to obtain the **prime factors** of this number. For example, prime factors of 24 are 2, 2, 2, and 3, whereas prime factors of 35 are 5 and 7.
7. Write a function which receives a float and an int from main(), finds the product of these two, and returns the product which is printed through main().
8. Write a function which receives 5 integers and returns the sum, average, and standard deviation of these numbers. Call this function from main() and print the results in main().
9. A 5 digit positive integer is entered through the keyboard. Write a function to calculate sum of the digits of the number
 1. Using recursion
 2. Without using recursion
10. A positive integer is entered through the keyboard. Write a program to obtain the **prime factors** of the number. Modify the function suitably to obtain the prime factors recursively.
11. Write a recursive function to obtain the first 25 numbers of a **Fibonacci sequence**. In a Fibonacci sequence, the sum of two successive terms is the third term. Following are the first few terms of the Fibonacci sequence:

1 1 2 3 5 8 13 21 34 55...